EVENTUAL PERIODICITY OF LINEAR GENERALIZED STONE–WALES PATHS IN AFFINE FULLERENE FAMILIES WITH CONVEX LATTICE-FACE DECOMPOSITIONS

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Abstract. We study literal linear generalized Stone–Wales paths in fullerene duals. For an integer affine family with a fixed decomposition into embedded convex lattice polygons in the six triangular-lattice directions, we give a uniform reduction of literal-path existence to a fixed monadic second-order sentence over unary finite words. The construction uses maximal face passages, their two-row outputs and all vertex collisions. Eventual periodicity follows from the known equivalence between MSO and regular languages. The result is a geometric definability reduction. We apply it, together with separate arithmetic arguments, to eleven specified catalogue families. The specification-level periodicity theorem is Lean-proved, with a prose correspondence lemma connecting its assumptions to the geometric hypotheses. Family-specific Lean proofs establish the K3, C1 and D12 path classifications, O2 absence on its stated domain, and the E10 classification for $r \geq 2$. Positive-ray actual-path periodicity is Lean-proved for K3, C1, O2, D12, D14 and E10, including the E10 boundary $r = 1$. We also describe closed conditions, exact decisions on two example rays for every ray parameter, finite checks, and a fixed-oblique-direction extension as a separate proof sketch.

1. Introduction

A literal linear generalized Stone–Wales path is a simple alternating triangle strip between degree-five vertices of a fullerene dual [2]. In an affine family, both the graph size and the number of face passages can grow with the parameter. The question is whether path existence nevertheless becomes periodic on an integer ray.

Our main construction represents passages and every possible surface-vertex collision in one fixed finite-word MSO sentence. Theorem 1a treats embedded convex lattice faces with fixed affine gluing. Corollary 1b applies to eleven specified families from Graver's catalogue [3]; A3 uses its separate classification and arithmetic argument [1]. Section 11.3 places this geometric reduction beside the existing logical and fixed-boundary periodicity theorems.

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2. What is proved, and how

Claim	Present proof basis	Scope
Theorem 1a	Lean specification theorem; prose geometric correspondence	Convex six-direction disk decompositions, with fixed affine gluing and eventual validity; fixed-residue variants.
Logical periodicity step	External MSO/regular-language theorem	One fixed sentence over unary finite words.
Corollary 1b	Geometric decompositions and arithmetic; direct Lean family proofs	Eleven specified valid positive-parameter families.
K3, C1, O2, D12, D14, E10	Lean quotient-graph proofs	K3 iff for $r, s \geq 2$; C1 iff for $a, b \geq 1$; D12 iff for $r, s \geq 1$; O2 absence for $t, d \geq 2$; E10 iff for $p, s \geq 1, r \geq 2$; all six positive-natural ray results, including E10 $r = 1$.
Family boundaries	Lean explicit paths and finite checks	O2 paths on $(1, d)$, $d \geq 3$, and $(t, 1)$, $t \geq 3$; D14 paths on $(1, s)$, $s \geq 2$, and $(r, 1)$, $r \geq 2$. Small remaining boundary decisions are finite evidence.
Other closed conditions and examples	Prose geometry, Lean predicate arithmetic, explicit witnesses and checkers	The domains and finite ranges specified in Sections 7, 8 and 11.2; C1/O2 IPR is prose.
D12 classification, further proofs	Two prose/computer-assisted proofs, in addition to the Lean proof	Positive integer r, s, with the quotient, geometric and certificate semantics stated in Section 12.
Fixed-oblique extension	Proof sketch	The four transfer obligations in Section 10.

The formal family claims concern the explicit quotient models; their catalogue identification uses the stated coordinate conventions. The convex-face result has two separately developed derivations on the common six-direction domain. Human proof peer review and the priority comparison remain open. The supplementary `VERIFICATION-RECORD.md` records provenance, implementation identifiers, independence qualifications and validation history.

3. Definitions, signatures and valid rays

3.1. The literal path.

Definition 1. In the dual G of a fullerene, a literal linear gSW path is a sequence $p_0, \ldots, p_e$ with odd $e \geq 3$, all vertices distinct, consecutive and distance-two vertices adjacent, $\deg p_0 = \deg p_e = 5$, and $\deg p_1 = \deg p_{e-1} = 6$.

Machine-readable claim: `def:literal`.

This is Definition 1(a) of [2]. Internal degree-five vertices are allowed. Even vertex count is part of the definition, not a new obstruction. Positive means existence of such a path; no chemical barrier, radial operation, or operation permitting heptagons is inferred. A fullerene is a simple three-connected cubic plane graph with only pentagonal and hexagonal faces. Its dual is a spherical triangulation with twelve degree-five vertices and all other degrees six.

When every three-cycle is facial and an edge has just the two common neighbours supplied by its incident triangles, an oriented triangle (a, b, c) has a unique next vertex: the common neighbour of b, c different from a. A simple path is a prefix of this forced sequence. If ρ is its first repeated-vertex index, a legal endpoint must occur strictly before ρ, not at ρ.
Machine-readable claim: `prop:forced`.

For fullerene duals, cyclic edge-connectivity [4] excludes nonfacial three-cycles and supplies the two-common-neighbour property. The finite-graph decision argument applies once this local geometric hypothesis and the start are fixed.

3.2. **Signatures and eventual validity.** *Machine-readable claim:* `def:signature`.

Work in the triangular lattice $\Lambda = \mathbb{Z}^2$, with $R(x, y) = (-y, x + y)$. A signature specifies a fixed finite set of polygon charts, paired boundary segments, corner equivalence classes, and transition maps $z \mapsto R^j z + d$. Corner coordinates and translations are integer affine functions of the parameters; rotations and gluing combinatorics are fixed. Along a ray $p(t) = at + b$, $t \in \mathbb{N}$, we require eventually positive edge lengths, embedded chart interiors, only the specified seam identifications, noncrossing macro edges, and a valid simple fullerene dual. Every cone has a marked corner label. Seams need an ownership convention, and full corner equivalence classes are used. Positive parameters and positive area alone do not prove validity.

A component $a_i t + b_i$ eventually lies in the positive integers exactly when $a_i > 0$ or $a_i = 0, b_i \geq 1$. In the first case an entry bound is $\max(0, \lceil (1 - b_i)/a_i \rceil)$. Replacing 1 by 2 gives the analogous bound for a lower limit two. A negative slope has no infinite positive tail. These arithmetic entry bounds do not certify the quotient.

4. The main result: convex lattice-face decompositions

Theorem 1a (Convex lattice-face rays)**.** *For all sufficiently large integers t, suppose a valid simple fullerene dual has a decomposition into a fixed finite number of embedded closed convex lattice-polygon disks. Every side has positive length and one of the six lattice directions. Straight angles and auxiliary degree-six corners are permitted. No distinct boundary points of one disk are identified. Corner coordinates and transition translations are integer affine in t, while rotations, side pairings and corner identifications are fixed. All degree-five vertices occur among the marked corners, with a fixed label set. Then the set of t for which a literal path exists is eventually periodic. The conclusion also holds when these hypotheses hold separately on finitely many residue classes modulo a fixed integer.*

Machine-readable claim: `thm:aligned`.

Thus some $t_0 \geq 0, P \geq 1$ satisfy the same decision at t and $t+P$ for all $t \geq t_0$. This is an existence theorem for a threshold and positive period. The following correspondence lemma connects the geometric hypotheses to the specification-level theorem, while Section 5 gives the passage-to-word proof.

Prose correspondence lemma. The geometric hypotheses of Theorem 1a determine a fixed `Sig` S and a tail on which `Valid` $S\,t$ and `TrianglesInCharts` $S\,t$ hold; the quotient `graph` $S\,t$ is the stated fullerene dual and `LiteralPath` is Definition 1.

Prose justification. For a convex lattice polygon with the six permitted side directions, its supporting lines give exactly three closed strips $L_x \leq x \leq U_x$, $L_y \leq y \leq U_y$, $L_s \leq x+y \leq U_s$. Their extrema are selected from finitely many affine corner coordinates; after their orders stabilize, all six bounds are integer affine. Retain the fixed charts, rotation exponents, affine translations and marked corners as the fields of S. For each paired side, use its prescribed transition as a gluing record.

No extra identifications from unrestricted records. Let g unfold a paired segment $A \subset K$ onto $g(A) \subset K'$, and let ℓ be its supporting line. The consistently oriented disks $g(K)$ and K' occupy opposite closed supporting halfplanes, so $g(K) \cap K' \subset \ell$. Their overlap there is an interval I containing $g(A)$. If I continues beyond an endpoint of $g(A)$, both supporting edges continue straight at that point. The two disks each contribute angle π. A fullerene dual has total cone angle $5\pi/3$ or 2π, so the point is flat, the two disks exhaust its neighbourhood, and their actual seam must continue across it. The orientation-preserving lattice isometry along the continued seam is the same g: it agrees on a nontrivial line segment. Repeat across the finitely many side subdivisions

until reaching the endpoint of I. Thus every pair identified by this unrestricted record is truly the same surface point, even if it lies beyond the named subdivision. Conversely, the prescribed paired sides are included in the records. A self-pairing of distinct points is excluded by the embedded-closed-disk hypothesis; an identity record is harmless. Transitive closure therefore gives exactly the prescribed seam and full corner equivalences, without extra identifications. This supporting-line and cone-angle argument is a prose geometric argument. Thus sites modulo these records are precisely the surface vertices, and common-chart unit steps are precisely the graph edges. Closed-disk embeddedness supplies the injectivity clause of `Valid`; the fullerene degrees and marked-corner hypothesis supply its degree clauses.

Use the same external fullerene fact as in Section 3: all dual three-cycles are facial, by cyclic edge-connectivity [4]. Each edge therefore has exactly the two distinct opposite vertices of its incident triangular faces as common neighbours. This gives `TwoCommonNeighbors`. Since chart boundaries follow lattice edges, no open unit triangle is cut by a boundary; the covering decomposition puts every triangular face inside a closed chart. Every graph triangle is facial, so its three vertices lift there with all three differences unit vectors. This is exactly `TrianglesInCharts`. These geometric identifications and the external fullerene fact supply the geometric correspondence.

Apply the specification-level periodicity theorem on the resulting tail. For fixed residue branches write $t = r+qn$ and apply the construction to each branch. The residue-class theorem combines the branches in the original parameter, assuming eventual agreement of the decision with each branch's `PathExists`. That agreement is supplied by this prose representation lemma; the combination of eventual periods is Lean-proved.

Corollary 1b (Specified catalogue families)**.** *On every integer affine ray eventually remaining in the valid positive integer domain of each of A1, A2, B2, C1, K3, E10, D12, D14, D1 and O2, literal-path existence is eventually periodic. The same conclusion holds for A3 on rays eventually satisfying $k > l > 0$, by its separate classification and arithmetic argument* [1].

Machine-readable claim: `cor:catalog`.

There are eleven families in this statement. Positive boundaries with a parameter fixed at 1 are included, notably K3 and O2 and E10 with $r = 1$. Collapsed zero boundaries are not. The corollary gives no two-variable closed classification merely from ray periodicity. The K3, C1, O2, D12 and D14 parts have direct family-specific ray proofs on their quotient graphs, including positive boundary parameters. E10 also has a formal actual-path ray proof: its $r = 1$ branch applies the formal general periodicity theorem with the validity and triangle-lifting hypotheses proved on the E10 model. These six results use no general prose correspondence lemma. Their ray declarations start with positive natural coordinates and nonnegative slopes; an eventually positive integer ray is covered by reindexing its positive tail. The other family routes retain their stated scope. A3 generally has changing parent-edge directions and is not silently placed under Theorem 1a. The catalogue terminology and corrections are Graver's [3]; the constructions are not claimed as new. Additional leapfrog families are discussed only as prospects in Section 10.

5. Proof of the convex-face theorem

The proof uses the common six-direction domain and its restricted arithmetic-to-word reduction. The fixed-oblique-direction sketch is treated separately in Section 10.

5.1. Maximal face passages and their outputs.

Machine-readable claim: `lem:rows`.

Rotate a passage direction to $(1, 0)$. A suitable phase of the forced vertex sequence is

$$q_{2j} = (j, h), \qquad q_{2j+1} = (j, h + 1). \tag{1}$$

A convex six-direction polygon is specified by

$$L_x \le x \le U_x, \qquad L_y \le y \le U_y, \qquad L_s \le x + y \le U_s.$$

Its bounds are affine in t after the finitely many max/min orders stabilize. Redundant collinear sides do not alter this description. With $L_y \le h \le U_y - 1$, the triangle $T_i = \{q_i, q_{i+1}, q_{i+2}\}$ lies in the polygon precisely in the ranges

$$\begin{aligned} A_e &= \max(L_x, L_s - h), & B_e &= \min(U_x - 1, U_s - h - 1), \\ A_o &= \max(L_x, L_s - h - 1), & B_o &= \min(U_x - 1, U_s - h - 2). \end{aligned} \tag{2}$$

Even indices are $2j$ for $A_e \le j \le B_e$, odd indices are $2j+1$ for $A_o \le j \le B_o$. Put $\delta = A_e - A_o$ and $\epsilon = B_o - B_e + 1$, both in $\{0, 1\}$. The full triangle-index set is empty or the interval

$$\ell = 2A_e - \delta \le i \le u = 2B_e + \epsilon. \tag{3}$$

This includes the one-odd-triangle case $A_e = B_e + 1$, $\delta = \epsilon = 1$. It must not be discarded as an empty even range.

The first two vertices of an entered passage already belong to the previous passage. Its new output is only $q_{\ell+2}, \dots, q_{u+2}$, comprising

$$\begin{aligned} y = h: &\quad A_e + 1 \le x \le B_e + 1, \\ y = h + 1: &\quad A_e + 1 - \delta \le x \le B_e + \epsilon. \end{aligned} \tag{4}$$

For example the upper row has $x \ge L_x + 1 - \delta$, $x + y \ge L_s + 2 - \delta$, $x \le U_x - 1 + \epsilon$, and $x + y \le U_s + \epsilon$. Empty rows are allowed, but a nonempty passage has nonempty output.

Root and terminal passages can be truncated. There are finitely many marked cone corners and at most ten oriented initial triangles at each. The initial triangle fixes h and an index of the form twice an affine expression plus fixed parity. Store its first two vertices separately. An endpoint corner and choice of output row similarly determine the last index. Truncation adds affine bounds on x. Both cuts are made when the whole path lies in one passage. Internal degree-five corners are not compulsory stopping points. The two endpoint neighbours must be distinct from all degree-five labels.

5.2. **States, transitions and all collisions.** *Machine-readable claim:* `lem:restricted`.

A state is (τ, h), where the finite type τ records face, direction, max/min branches, parity, subdivided entry/exit side, and root/terminal information. A constant C bounds $|h| \le Ct + C$. The number of states or face passages need not be bounded independently of t. The relevant relations are Boolean combinations of

$$\pm h \le at + b, \qquad \pm h \pm h' \le at + b, \qquad at + b \ge 0, \tag{5}$$

and fixed-modulus congruences. A relation such as $2h + h' \le at + b$ with distinct free variables is not part of this language. Substituting one variable for another can be reduced by fixed parity and rounding.

Within a fixed type, each endpoint's three normal coordinates has the form $c(t)$, $h + c(t)$ or $-h + c(t)$. When the last ordered pair of a passage is glued to the next one's first ordered pair, the directions are rotated together. In direction-aligned coordinates the linear part of this transition is the identity, so $h' = h + c(t)$, not a doubling map. The paired unit edge determines the opposite face; its interior contains no macro corner. The conditions therefore have form (5).

Each output row fixes one of $x, y, x + y$ at $\pm h + c(t)$ and imposes affine bounds in all three normals. For distinct row normals the determinant is ± 1: their intersection is integral, and the third normal is their sum or difference. Substitution into both rows gives (5). Equal normals first require equal values; the remaining integer interval is nonempty exactly when every lower bound is at most every upper bound. The same restricted form results. Between different faces, compare through every shared side and every common marked corner label. Rotations permute the three normals with signs. Embeddedness and fixed gluing ensure these exhaust quotient identifications, even when faces share several sides. Auxiliary flat corners are included. Comparison with the root's

two saved vertices uses the same formulas. Since an embedded face row is injective, pairwise disjoint outputs and their disjointness from the root are exactly the full vertex-distinctness condition.

5.3. **Folding the restricted arithmetic into a finite word.** *Machine-readable claim:* `lem:folding`.

For $t \geq 2$, write a height variable uniquely as $h = jt + y$, $0 \leq y < t$. The quotient j ranges over a fixed finite set. Set $t = 2n + \eta$, $\eta \in \{0, 1\}$, and split y into

$$L : y = v, \qquad R : y = t - 1 - v \quad (0 \leq v < n), \qquad M : y = n \quad (\eta = 1). \tag{6}$$

Original type, j and $L/R/M$ are finitely many copies of the positions of a unary word of length n. A middle copy is placed at the first position and interpreted as coordinate n, independent of that position.

For two ordinary positions the inequalities become

$$\alpha v + \beta w \leq 2cn + d, \qquad \alpha, \beta \in \{-1, 1\}.$$

The coefficient of n is even. If $\alpha = -\beta$, then for $c \neq 0$ truth stabilizes, while $c = 0$ is order with a fixed successor offset. If $\alpha = \beta = 1$, the sum lies between 0 and $2n - 2$. The only nontrivial cases are $c = 0$, a finite list near the left end, and $c = 1$. For $c = 1$, putting $a = n - 1 - v, b = n - 1 - w$ gives $a + b \geq -d - 2$, either automatic or the exclusion of finitely many near-end pairs. The case $\alpha = \beta = -1$ is its integer complement. If one position is a middle copy, only one variable remains and the nontrivial inequalities are fixed distances from an end. Two middle copies give an affine truth value that stabilizes.

Fixed-modulus congruences are represented by cyclic MSO colours on the word: the first position has colour zero and each successor increments the colour. The final colour determines n modulo the fixed modulus. Finite copies of element or set quantifiers are expanded copy by copy. Thus every fixed Boolean combination used above is MSO-definable. This argument does not permit an arbitrary Presburger relation such as $v + w \leq n$; it uses the even coefficient obtained by the folding step.

5.4. **The exact path sentence and unary regularity.** *Machine-readable claim:* `lem:mso-path`.

Choose a nonempty state set S, root a and terminal b. Terminal states have no outgoing transition and root states no incoming one. Record the parity of the number of vertices already output as a finite type. Initially it is even, since the root has two saved vertices; each transition adds the parity of $u - \ell + 1$. Impose the following MSO conditions:

(1) If $a = b$, $S = \{a\}$. Otherwise a has in/out degrees $(0, 1)$, b has $(1, 0)$, and every other state has $(1, 1)$ within S.
(2) Every nonempty proper subset of S has a transition, in either direction, across its boundary. This is unary-set connectivity.
(3) Outputs of distinct states are disjoint, and all outputs are disjoint from the two saved root vertices.
(4) Endpoint degrees are $5, 6, 6, 5$ as in Definition 1. Total vertex count is even and at least four.

For one state, the last size condition is $u - \ell + 1 \geq 2$; with two or more nonempty-output states it is automatic. Degree conditions are membership or nonmembership in the finite marked-cone set. No binary relation quantifier is used.

On a finite set, the first two conditions give one directed path through all of S: a reached cycle would create an extra incoming edge, and a separate cycle would contradict connectivity. Join the passages in this order. Ordered seam pairs give consecutive and skip adjacency; the third condition gives distinct vertices. Size, parity and degree conditions give exactly the literal path.

Conversely, split any literal path into maximal consecutive passages in its convex faces, truncating only at its ends. Equations (2)–(4) recover their states. Nonempty output and path simplicity

prevent a repeated state. Any extra eligible successor within S would share the next output vertex with the actual successor, contradicting disjointness. Hence the selected states satisfy all four conditions. One-passage paths and paths with internal cones are included in both directions.

Finite-word MSO defines exactly the regular languages [5, Section 3.1, Theorem 3.2]. Apply this established result only at this last logical step. For each η, a deterministic automaton over one letter eventually repeats a state, so its accepted lengths n are eventually periodic. Combine $t = 2n + \eta$ and, if necessary, the initially fixed residue branches. This proves Theorem 1a. The geometric passage construction supplies the fixed sentence to which the external automata theorem applies.

6. Applying the theorem to the specified families

Machine-readable claim: `prop:family-decompositions`.

D12 uses two subdivided hexagons and D14 two convex hexagons whose long sides are subdivided with lengths r, s, r. O2 uses two hexagons and six triangles. K3 uses two caps, each divided into four triangles, and three hexagons. These decompositions retain the positive parameter-1 boundaries.

D1's original thirty-degree sides do not directly satisfy the six direction hypothesis. Cut a triangle $A_iB_iW_i$ from each oblique side and move it to the back by the same lattice isometry. For example,

$$A_0 = (r, 0), \quad B_0 = (r + p, p), \quad W_0 = (r, p), \quad C = (-p, r + 2p),$$

and $W_0 - C = \alpha(A_0 - C) + \alpha(B_0 - C)$ with $\alpha = (r + p)/(2r + 3p)$ and $0 < 2\alpha < 1$. The pieces are contained and disjoint. The result is eight convex pieces: one regular hexagon, six quadrilaterals, and the back hexagon. New corners are flat degree six. Area equality alone is not used to infer the same quotient.

For E10 put $m = 2r + s$, tube circumference $C = 3m$. Move the longitudinal cut from (p, p) to $(0, 2p)$, obtaining three lattice parallelograms of width $m < C$ and two caps: five disks, also when $r = 1$. Cut-corner positions involve $p \bmod C$. If C grows, $\lfloor p/C \rfloor$ eventually stabilizes; if C is constant, use fixed residue classes. Delete duplicate auxiliary corners and fix their order to obtain affine coordinates and fixed gluing. This does not classify E10 with $r = 1$.

Family and domain	Reason for all valid affine rays
A1, $r \geq 1$	Integer GC absence, including the separate $r = 1$ case; eventual period one.
A2, $p \geq 1$	Nonconstant rays eventually give nonprimitive GC images; constant rays are constant graphs.
B2, $p, r \geq 1$	Its closed condition is a finite disjunction of affine equalities.
C1, $a, b \geq 1$	Lean graph iff $b = 1$ and actual-path ray periodicity; no general prose bridge.
A3, $k > l > 0$	Separate complete classification and arithmetic; Section 8.
K3, $r, s \geq 1$	Lean literal-path ray periodicity, including parameter 1; no general prose bridge is used.
E10, $p, r, s \geq 1$	Lean actual-path ray periodicity; for $r = 1$, the formal general theorem applies with model hypotheses proved in Lean.
D12, $r, s \geq 1$	Lean actual-path ray periodicity, including parameter 1; no general prose bridge.
D1	The convex decomposition on valid positive integer parameters.
D14, $r, s \geq 1$	Lean actual-path ray periodicity, including parameter 1; no general prose bridge.
O2, $t, d \geq 1$	Lean actual-path ray periodicity, including parameter 1; Lean absence for $t, d \geq 2$. No general prose bridge.

These decompositions and the separate arithmetic results prove Corollary 1b on its stated domains.

7. Closed conditions and the scope of partial results

The following closed conditions concern valid catalogue graphs on the displayed domains.

7.1. **The specified closed domains.** *Machine-readable claim:* `prop:closed`.

Family/domain	Literal path condition	Proof and domain
A1, $r \geq 1$	Never positive	Integer GC for $r \geq 2$; at 1 no degree-six vertex.
A2, $p \geq 1$	$p = 1$	C_{60} witness and integer GC.
C1, $a, b \geq 1$	$b = 1$; IPR iff $a, b \geq 2$	Graph iff in Lean; IPR equivalence by prose geometry.
B2, $p, r \geq 1$	$p = 1$ or $\lvert p - r\rvert = 1$	Prose proof on the whole positive domain.
K3, $r, s \geq 2$	$s \equiv \pm 1 \pmod{r}$	Graph iff in Lean on the displayed domain.
O2, $t, d \geq 2$	Never positive	Graph absence in Lean; IPR by prose geometry.
E10, $p, s \geq 1, r \geq 2$	$m = 2r + s$ and $p \equiv d \pm 1 \pmod{m}$ for some $d \in \{0, r, r + s\}$	Graph iff in Lean on the displayed domain; the ray theorem also covers $r = 1$.

K3 writes $s = qr + u$: $u = 0$ is excluded by integer GC, $2 \leq u \leq r - 2$ by belts of widths $r, u, r - u$, and $u = 1, r - 1$ by explicit simple paths. The overlapping small cases $r = 2, 3$ are separate. The iff is not extended to $r = 1$ or $s = 1$. E10's candidate congruences come from the positions of corners around the tube; reaching a cone does not alone establish simplicity. The E10 argument uses eighteen parameter regions, their walls and explicit witnesses.

7.2. **Partial branches and boundary examples.** *Machine-readable claim:* `prop:partial`.

For O2, absence holds on $t, d \geq 2$, with IPR on that domain supplied by prose geometry. Boundary paths exist at O2$(1, d)$ for $d \geq 3$ and O2$(t, 1)$ for $t \geq 3$. At O2$(1, 1)$, O2$(1, 2)$ and O2$(2, 1)$, the path decisions have finite computational evidence. Every constant graph has a constant ray decision, independently of computing it. For D14, absence holds for $r, s \geq 2$, and explicit paths exist at $(1, s)$ for $s \geq 2$ and $(r, 1)$ for $r \geq 2$. These are Lean graph lemmas; the D14 ray theorem includes all positive boundary rays. D14$(1, 1)$ has a finite path witness. For D1 with both parameters at least two, a belt-based absence argument is a proof sketch. Common integer scaling by at least two gives absence in the specified leapfrog domains, without classifying their primitive points.

For E10$(p, 1, s)$, $m = s + 2$, explicit witnesses give sufficiency when $p < m$ or $p \bmod m \in \{0, 1, 2, m - 2, m - 1\}$. General necessity is not proved. Among the 84 positive (p, s) with $N = 12 + 2s^2 + 12s + 12p(2 + s) \leq 740$, exactly E10$(9, 1, 4) = C_{740}$ is negative; all 83 smaller examples are positive. E10$(17, 1, 11)$ and $(31, 1, 7)$ give further negatives at C_{3038}, C_{3542}. These three examples are non-IPR. Exhaustive construction and independent checks establish the stated minimality inside the $r = 1$ E10 family.

For D12, the following partial branches apply: $\gcd(r, s) \geq 2$ gives absence; $r \geq 3$, $s = qr \pm 1$, $q \geq 1$ gives existence by the separately derived simple witnesses. The additional $\lvert r - 3s\rvert = 1$ and $r \geq 4s$ branches are construction and absence arguments with their stated geometric conditions. The full D12 classification, with a Lean proof, is given in Section 12.

8. Arithmetic on affine rays and exact examples

8.1. **The determinant reduction.** *Machine-readable claim:* `lem:divisibility`.

Let $A(t) = \alpha t + \beta$ be eventually positive and $B(t) = \gamma t + \delta$. A constant A gives a fixed-modulus congruence. If $\alpha > 0$, then

$$A \mid B \quad \Longrightarrow \quad A \mid \alpha\delta - \gamma\beta. \tag{7}$$

For nonzero right side, divisibility is impossible once $A > |\alpha\delta - \gamma\beta|$. For zero right side, the ratio $B/A = \gamma/\alpha$ is constant and is either always integral or never integral. Finite Boolean combinations of affine divisibilities are eventually periodic; a growing modulus gives eventual constancy. Take a common multiple of fixed moduli and a maximum stabilization threshold. This elementary arithmetic is not proposed as new number theory. The following exact predicates have Lean arithmetic proofs.

For all integer ray coefficients, define parameters by $(at + b).\,\mathrm{toNat}$. The formal statements for all rays use *domain and positive predicate*; they do not call out-of-domain graphs positive. On an eventually positive domain the conversion agrees with the integer affine parameter.
Machine-readable claim: `thm:k3-arithmetic`.

For K3, eventual periodicity holds for K3Dom $\wedge$ K3Pos on all integer affine rays, where the domain is $r, s \geq 2$ and the predicate is $r \mid (s-1)$ or $r \mid (s+1)$. If r is nonconstant on a valid tail, the predicate is eventually constant. If $r = b \geq 2$ is constant, period b is valid from any specified domain-entry threshold. These need not be least periods.
Machine-readable claim: `thm:e10-arithmetic`.

For E10, the corresponding all-ray statement holds for domain $p, s \geq 1, r \geq 2$ and the six divisibilities in the E10 condition. If $2r + s$ is nonconstant, the predicate is eventually constant; when it is constant, that positive modulus is a period on the given domain tail.
Machine-readable claim: `thm:other-arithmetic`.

For B2, C1, A1 and A2, eventual constancy holds for the domain-qualified predicates. The conjunction/disjunction/negation closure and domain qualification are proved in the same arithmetic layer.

8.2. **A3 is a separate variable-direction case.** *Machine-readable claim:* `thm:a3-arithmetic`.

For $k(t) = at + b, l(t) = ct + d$ set $\Delta = ad - bc$. If $\Delta \neq 0$, $\gcd(k, l) = \gcd(k, l, \Delta)$, so primitivity is periodic modulo $|\Delta|$. When $\Delta = 0$ and the ray is nonconstant, it is eventually an increasing integer multiple of a fixed primitive direction and hence negative by integer GC. Constant rays are constant. In the primitive case put $m = \min(l, k - l)$. The A3 criterion is $m = 1$, or quotient $q = \lfloor k/m \rfloor \geq 3$ with $m \mid (k \pm 1)$, or $q = 2$ with $r = k - 2m = 1$ or $r \mid (m \pm 1)$. Branches and quotient stabilize, including an integer limiting ratio by checking the sign of the constant remainder. The persistent $q = 2, r = 0$ growing branch is not primitive. Apply (7) to the remaining tests.

The first paper proves the full graph classification and a graph-path ray theorem. Its arithmetic proof establishes a threshold and positive period; an independent arithmetic derivation permits period $|\Delta|$ for nonconstant $\Delta \neq 0$ rays and period 1 otherwise. These are valid periods, with no claim of leastness. After classification, eventual periodicity is also a consequence of [6, Theorem 1.15].

8.3. **Four previously solved rays.** *Machine-readable claim:* `ex:exact-rays`.

Ray, $t \geq 0$	Positive values	Valid period and threshold
$\mathrm{D12}(124 + 3t, 41 + t)$	All t	$P = 1, t_0 = 0$
$\mathrm{D12}(125 + 3t, 41 + t)$	None	$P = 1, t_0 = 0$
$\mathrm{K3}(43 + 2t, 170 + 5t)$	$40, 42$	$P = 1, t_0 = 43$
$\mathrm{E10}(200 + 5t, 17 + 2t, 23 + t)$	$3, 10, 17, 34$	$P = 1, t_0 = 35$

Compressed-word calculations and independent family-condition arguments give these exact all-parameter decisions.
Machine-readable claim: `ex:k3-formal`.

For every $t \in \mathbb{N}$, the K3 arithmetic theorem gives $\mathrm{K3Pos}(43 + 2t, 170 + 5t) \leftrightarrow (t = 40 \vee t = 42)$.
Machine-readable claim: `ex:e10-formal`.

The E10 arithmetic theorem gives the corresponding predicate iff $t \in \{3, 10, 17, 34\}$. The determinant constants used are $123, 127$ for K3 and $142, 144, 739, 749, 681, 691$ for E10. Divisibility

bounds t before the remaining cases are checked by kernel arithmetic. These are all-t arithmetic proofs, not graph iff proofs by themselves.

For every $s \geq 41$, D12$(3s+2, s)$ is negative and D12$(3s+1, s)$ has a path with $56s+12$ distinct vertices. The computer-assisted proof checks 2,058 nonnegativity conditions and 756 no-integer-solution conditions, 2,814 in total. A separate construction at $s = 58$ gives a 3,260-vertex positive witness.

9. Specification and family-specific graph proofs

For every fixed signature S and t_1, the specification-level theorem states that if `Valid` $S\,t$ and `TrianglesInCharts` $S\,t$ hold for every $t \geq t_1$, then some t_0 and positive P satisfy `PathExists` $S\,t \leftrightarrow$ `PathExists` $S\,(t+P)$ for all $t \geq t_0$. These assumptions require the local validity and triangle-lifting properties; the geometric correspondence lemma supplies them from the convex disk decomposition. The residue-class variant combines fixed branches after their decisions agree with the original family.

9.1. K3 graph classification and ray periodicity.

Machine-readable claim: `thm:k3-sufficient`.

For natural $r, s \geq 2$, K3 has a literal path if and only if $s \equiv \pm 1 \pmod{r}$. A classification of graph triangles in the cap or tube proves that no edge has three common neighbours and supplies the forced-continuation argument for necessity.

Literal-path existence is eventually periodic on every ray $(r, s) = (r_0 + at, s_0 + bt)$ with $r_0, s_0 \geq 1$ and $a, b \geq 0$. The proof constructs the common-neighbour property directly. Nonconstant parameter-one boundary rays use explicit eventual paths; constant rays have constant decisions. The modular iff has domain $r, s \geq 2$.

The specification uses eleven convex polygons, equivalence closure of explicit affine side gluings, and adjacency from unit differences inside a common chart. The literal path and actual neighbour-set degree definitions are those of the first paper's specification.

The proof uses a tube construction. A glued-site invariant is $(\mathrm{hgt}, \mathrm{ang} \bmod 6r)$. It is preserved by all 42 directed edge gluings. It is not globally injective: the two central cap triangles are not separated by it. On tube points with horizontal coordinate difference at most one, however, equality of quotient vertices forces equality of points. The explicit paths are

$$P_n^+ = (-(n \bmod 2), \lfloor (n+1)/2 \rfloor), \qquad P_n^- = (n \bmod 2, \lfloor n/2 \rfloor),$$

with $4s-2$ vertices for $s = qr+1$ and $4s+4$ for $s = qr-1$. Chart membership, consecutive and skip adjacency, distinctness, and the four endpoint degree requirements are proved on the quotient. The invariant is not used to claim a global vertex normal form.

Machine-readable claim: `lem:k3-corners`.

All twelve labelled corner representatives have degree five. These representatives supply the endpoint degree-five witnesses.

Machine-readable claim: `lem:k3-hex-interior`.

Hexagon interiors have degree six.

Machine-readable claim: `lem:k3-tri-interior`.

The six noncentral cap-triangle interiors have degree six. The necessity argument uses the cap/tube triangle classification and common-neighbour bound together with these local degree facts.

9.2. C1 graph classification and ray periodicity.

For C1, natural $r, s \geq 1$ admit a literal path if and only if $s = 1$. Here r, s correspond to the paper's a, b. Actual-path existence is eventually periodic on every ray $(r_0 + ut, s_0 + vt)$ with $r_0, s_0 \geq 1$ and $u, v \geq 0$, including constant rays. Eventually positive integer rays are covered by shifting their starting index to the positive natural domain.

The eight-cell specification has four hexagons with alternating sides s, r, s, r, s, r and four triangles of side r. A vertex bijection preserving adjacency identifies its quotient with the lattice surface in $\mathbb{Z}^4$ cut out by $0 \le x_i \le r+s$, $\sum_i x_i = 2r+s$, and some $x_i \in \{0, r+s\}$. The bijection preserves literal-path existence. The proof establishes degrees and excludes three common neighbours of an edge directly, then uses forced strips for $s \ge 2$ and an explicit path for $s = 1$.

9.3. **O2 absence and ray periodicity.** For natural $t, d \ge 2$, O2 has no literal path. Actual-path existence is eventually periodic on every ray $(t_0 + an, d_0 + bn)$ with $t_0, d_0 \ge 1$ and $a, b \ge 0$, including parameter-one and constant rays. Eventually positive integer rays are reindexed to their positive tail.

The eight-cell specification uses two hexagons with alternating sides $2t, d, 2t, d, 2t, d$, each long side split at its midpoint, and six triangles of side t. A graph equivalence identifies it with two overlaid hexagons sewn along the three d-sides, with a rhombic pocket inserted into each $2t$-side slit. The proof directly rules out three common neighbours on an edge, locates degree-five vertices at the twelve corners, and covers the 120 starting triangles by ten closed strips.

Explicit boundary paths are proved for $(1, d)$ with $d \ge 3$ and $(t, 1)$ with $t \ge 3$. Path existence at $(1,1), (1,2), (2,1)$ is supported by finite checks. These three constant graphs do not require their existence decision for the ray theorem: any fixed graph has a constant decision. Explicit finite witnesses give paths there.

9.4. **D14 ray periodicity and auxiliary boundary lemmas.** For D14, actual-path existence is eventually periodic on all natural rays $(r_0 + an, s_0 + bn)$ with $r_0, s_0 \ge 1$ and nonnegative slopes, including parameter-one and constant rays. Eventually positive integer rays are reindexed to a positive natural tail.

The specification consists of two hexagons with alternating side lengths $s, 2r+s$, subdividing each long side as r, s, r. A graph equivalence maps the quotient to two lattice hexagons whose boundary circles are identified with an offset between their corners. A seam vertex is recorded by its arclength modulo $6r + 6s$. The proof directly excludes three common neighbours on an edge and, for $r, s \ge 2$, locates degree-five vertices at the twelve corners. Twelve closed strips cover all 120 starts.

Absence holds for $r, s \ge 2$. Explicit paths exist at $(1, s)$ for $s \ge 2$ and $(r, 1)$ for $r \ge 2$. At $(1, 1)$, a finite path witness supplies the decision; the constant-ray statement follows directly from constancy of the graph.

9.5. **E10 graph classification and all positive rays.** For natural $p, s \ge 1$ and $r \ge 2$, the E10 quotient admits a literal path exactly under the six congruences in the closed-condition table. An isomorphism identifies the five-polygon quotient with two hexagonal caps and a tube of circumference $3m$ and height $2p$, where $m = 2r + s$ and the twist is $p \bmod 3m$. Degree five occurs exactly at the twelve cap corners. A line-state invariant excludes the six negative regions; explicit tube zigzags establish the positive residues.

Actual-path existence is eventually periodic for all positive natural starts and nonnegative natural slopes in p, r, s. When r is eventually at least two, the classification reduces the problem to affine divisibility. On a ray with $r = 1$, a five-chart affine signature has 75 gluing records and twelve marked corners. Its quotient graph is isomorphic to the tube model. Chart injectivity, degrees, the two-common- neighbour property and triangle lifting are proved on that model, supplying the hypotheses of the formal general periodicity theorem. Residue classes make the twist affine, and their eventual periods combine. This formal route leaves the separate $r = 1$ sufficient conditions above and their open converse unchanged; it gives no $r = 1$ closed classification.

9.6. **Conditional auxiliary lemmas.** *Machine-readable claim:* `lem:historical-kernels`.

The auxiliary lemmas have the following hypotheses: `finite_literal_path_decision` assumes a finite type and two common neighbours; GC lattice transport excludes odd endpoints in a scaled

lattice; the four cylinder arithmetic declarations prove the height-kernel displacement with stated divisibility hypotheses; `induction_uniform_depth` assumes a finitely branching child list and genuinely decreasing lexicographic rank. The concrete `ietComponentChildren` models only removal of equal last labels, not every subtraction, equal-length elimination or accelerated suffix. The finite decision, scaled-lattice transport and rank statements are used with their displayed hypotheses.

10. An extension to fixed oblique directions (proof sketch)

Machine-readable claim: `claim:oblique`.

The proposed extension replaces the six side directions by an arbitrary fixed finite set of primitive lattice directions, keeping eventual validity, ownership and fixed affine gluing. For a direction v, use transverse coordinate $h = \det(v, z)$. Across a seam the shift is affine. Let D be a common multiple of the nonzero $|\det(v, e)|$ and put $Q = 4D$. Add the transverse cell modulo Q and t modulo Q to the finite labels. Entrance and exit indices are $k_{\text{in}} = \lceil \xi_{\text{in}} \rceil$, $k_{\text{out}} = \lceil \xi_{\text{out}} \rceil$ with $k_{\text{in}} \le k < k_{\text{out}}$. Reordering cells $n = \sigma + Qq$ conjugates the discrete midpoint permutation; it does not claim an order-preserving conjugacy of the continuous intervals.

The return-tower construction keeps a full partition into return towers, including saved periodic components. Equal final labels remove a periodic tower; equal lengths merge and remove a label without losing an orbit. Repeated winner cycles are compressed as powers. The claimed bound is on the number of Letter/Cat/Power nodes, not on expanded passage count. Base widths and exponents are affine on fixed residue branches; word lengths and displacement vectors are rational polynomials in t. A lexicographic measure uses leading coefficient sum in the original parameter, number of labels, and constant length; it is not silently reused after coefficient-amplifying reparameterization.

For local ownership, close the side directions under rotations. A unit triangle meets at most $\sum_e (\max_\Delta \det(e, z) - \min_\Delta \det(e, z) + 1)$ candidate macro-edge levels. Embedded noncrossing edges and finitely many corner displays bound the length of the local gluing word. A finite number of charts alone does not bound how often they can be revisited. Entrance-closed, exit-open ownership excludes repeated representations of one position from genuine quotient collisions. Keep the full corner class and the degree-six second-point test; non-IPR graphs can have fewer than 120 legal starts.

Tower return restores midpoint and direction; two returns restore left/right phase. Nonempty output, initial suffix and terminal prefix must be connected before using this as a finite first-repeat bound. Positions are leaf types and ancestor-power repetition indices. A free index j contributes j times a displacement depending only on t. Products jt or jt^2 are allowed coefficients in one-parameter Presburger arithmetic; products of two freely quantified repetition indices are not used. Fix one start address θ throughout:

$$\begin{aligned}\mathrm{Bad}(t, \theta, e) \iff \exists \xi, \eta\, [&D_\theta(\xi) \wedge D_\theta(\eta) \wedge \xi \le e \wedge \eta \le e \\ &\wedge \neg I_\theta(\xi, \eta) \wedge G_\theta(\xi, \eta)].\end{aligned}$$

Success would be $\exists \theta, e\, [\mathrm{Legal}(t, \theta, e) \wedge \neg \mathrm{Bad}(t, \theta, e)]$. Legal includes odd endpoint index, minimum length, degrees and the preceding output position. Allowing each collision a different start address would be incorrect.

The remaining obligations are: (i) write the first-crossed-edge, containment and order inequalities of the local geometric transfer, with an oblique-family example; (ii) verify the affine tower coefficients for the Q-split IET; (iii) keep effective threshold/period computation dependent on the effective elimination result [7]; (iv) compare an oblique-family ray's predicted tail against full graph decisions. These four conditions delimit the proof sketch. BGW supplies eventual periodicity only after the geometric formula has been established.

10.1. **Prospective applications only.** C2 and K4 are specified leapfrogs of C1 and K3; the identified $\mathrm{O1}(p, 0, s)$ subfamily is a specified leapfrog of O2. The local coordinate map $(x, y) \mapsto$

$(x - y, x + 2y)$ has determinant 3. These are prospects through the extension, not families in Corollary 1b. Leapfrog is not assumed to preserve absence in general, and the full three-parameter O1 family is not classified. These prospects do not cover arbitrary signatures or every degenerate boundary. For the prospective signatures, twice area is $24a^2 + 48ab + 12b^2$ for C2 and $42r^2 + 36rs$ for K4. Their $a = 0$ and $r = 0$ cases collapse faces, while $b = 0$ and $s = 0$ give zero sides. The zero middle coordinate in the identified O1 subfamily denotes a separately identified leapfrog signature, not permission to discard a zero edge in the old signature.

11. Fixed-width structure, finite checks and novelty

11.1. **Belts and a fixed cylinder.** *Machine-readable claim:* `prop:cylinder`.

Developed triangle midpoints lie on a straight line; vertices alternate on its two neighbouring integer-height rows. Under $z' = Mz + q$, a height $h(z) = n \cdot z + e$ transforms as $n' = nM^{-1}$, $e' = e - n' \cdot q$. If every start is covered by a return preserving position, direction and primitive height, no cone lies inside a band, and all widths are at least two, the required pair of endpoint rows cannot both occur. Width one alone is not sufficient: simplicity and endpoint-neighbour degrees remain necessary. For integer $\mathrm{GC}(k, 0)$, $k \geq 2$, the odd-point determinant ± 1 contradicts membership in the k-multiple lattice; rotations about parent vertices preserve that scaled lattice.

For the fixed-cylinder argument let circumference vector $W = gu$, $g = \gcd(|W_1|, |W_2|)$, $h(z) = \det(u, z)$, and insertion vector D have $H = h(D) > 0$. Let L be the least common multiple of the nonzero $|h(v)|$ over the six unit directions. An entrance-state period is

$$Q = 2gL, \qquad \frac{Q}{h(v)}(Hv - h(v)D) \in \mathbb{Z}W. \tag{8}$$

Directions with $h(v) = 0$ loop in the flat core and repeat without reaching a cone. If the caps and collars have a fixed finite vertex set C, at most $|C| + 1$ new cap visits occur before stopping. Overlap of the same three-vertex display is not a new visit. On a residue branch the passages have form $a_i(n) + jv_i$, $0 \leq j \leq \ell_i(n)$, with affine a_i, ℓ_i and fixed v_i, W. All collisions are $a_i(n) + jv_i = a_k(n) + qv_k + zW$, with distinct-position and range conditions. Ordinary Presburger arithmetic then gives periodicity. Q is an entrance period, not necessarily the final or least period.

The uniform cap argument does not extend to growing width/caps. The $\mathrm{D12}(3, 3t + 4)$ example has passage word $(BA)^{t+1}$ and first legal endpoint index $30t^2 + 88t + 59$. Its $2(t + 1)$ passages are unbounded.

11.2. **Finite checks and their domains.** *Machine-readable claim:* `sec:finite`.

An arithmetic comparison evaluates 1,809 values on nine known A3 rays, and fourteen full graph decisions use an independent constructor. The fourteen graph inputs are known examples and the arithmetic values form one sample set. Fixed tail coefficients beyond 64 are checked on those rays. The E10 descriptor audit checks eight charts, 36 inverse gluings, 108 intervals, 156 connections, 120 descriptors and twelve labels. A local polygon audit covers 71 charts, 342 edges and 21 single-zero boundaries. The return-tower audit has 6,564 inputs and 67,312 states. The phase audit covers 48 directions, 14,904 entrances, 4,005 intervals (1,013 empty), 5,496 lifts and 40,380 conjugacy cells.

An independent arithmetic implementation checks 378 coefficient representations and 13,430 pairs, with maximum period 44 and threshold 69 in those finite records. Passage-output tests cover 7,056 strips, 2,255 nonempty row outputs, 16,212 subintervals, 99,225 ordered row pairs and 509,160 folded-point comparisons. These finite ranges check the specified algebra and constructors.

Quotient-model mirror tests compare vertex and edge counts, degree sequences, seams and path decisions. C1 uses 36 pairs $1 \leq r, s \leq 6$; O2 uses 80 pairs $1 \leq t \leq 8$, $1 \leq d \leq 10$; D14 uses 81 pairs $1 \leq r, s \leq 9$. Statement-reader samples of 25, 36 and 25 pairs, respectively, are contained in those ranges. Generated degree and strip formulas are verified universally by their Lean proofs.

11.3. **Novelty and remaining obligations.** *Machine-readable claim:* `sec:novelty`.

Immediate known consequences. Once a fixed sentence ψ_η describes success on a^n, $t = 2n + \eta$, MSO/regular-language equivalence gives a finite automaton; its unary state orbit is eventually a cycle [5, Section 3.1]. Finite fixed-modulus residue branches preserve eventual periodicity. Likewise, a fixed-size parametric Presburger description gives the conclusion directly by [6, Theorem 1.15]. Neither theorem builds the geometric description: an unbounded list of path vertices is not a fixed list of quantified variables, and variable-variable products are not permitted by that Presburger theorem.

For a fixed-width construction $G_n = C[F^n(B)]$ with fixed finite boundary, literal-path existence has a fixed CMSO_2 description. Quantify a vertex set X, an edge set E' and distinct endpoints a, b: (X, E') is connected, has degree one at a, b and degree two elsewhere; $|X| \geq 4$ is even; a, b have ambient degree five and their path neighbours ambient degree six; every two-edge path in E' has adjacent outer vertices in the ambient graph. This permits interior cones and extra chords. The selected subgraph is precisely a literal path, in both directions. Finite-state recognizability for a fixed HR signature [9, Chapter 1, Theorems 1.12 and 1.21(2)] then makes iteration of F a finite-state orbit. Thus this fixed-width existence statement is a short known consequence, not a new principle. Mere bounded treewidth of an arbitrary indexed sequence is insufficient: the uniform repeated construction is essential. The explicit Q, cap-visit bound and actual period-calculation procedure require separate evaluation.

Nearby results and the missing translations. Behmaram–Boutillier's fixed-circumference matching transfer matrix has 2^m states [10, Section 3, Eq. (3.1)]; increasing width does not leave the same matrix, and its observable is matchings, not literal paths. Fischer–Makowsky prove periodicity or semilinearity of model-size spectra for CMSOL classes with bounded width [11, Theorems 1.4, 1.7 and 1.10]. Such a spectrum asserts existence of some model of a size, rather than deciding a specified G_t. Kotek–Makowsky's labelled (bi-)iterative graph families and MSO polynomials admit finite matrix transfers and C^2-finite recurrences [12, Definitions 17–20, Lemma 28, Theorems 30 and 32]. Their bounded-clique-width framework does not automatically include growing lattice patches; a recurrence for values is not itself Boolean periodicity. In each case the geometric family and its parameter require a uniform interpretation satisfying the cited hypotheses.

The syntax $\pm x \pm y \leq c$ is already the octagon domain [13, p. 1]. Bozga–Iosif–Konečný's difference, octagonal and finite-monoid affine relations have acceleration and Presburger transitive-closure results [14, Definition 1 and Sections 4.2.1–4.2.2]. These principally iterate a fixed relation and allow linear drift; here the relation itself varies with t, and all previously output surface vertices must remain distinct. A fixed-dimensional translation preserving that condition has not been supplied. The term "zigzag automata" there does not identify the subject with fullerene zigzags.

Jonoska–Krajcevski–McColm study counter languages of walks returning to a unit cell in periodic digraphs [15]; Wanke studies path/cycle algorithms on finite periodic graphs [16]. The available abstracts and bibliographic records describe these walk and algorithm problems. Walks or decidability alone do not preserve literal simplicity or prove periodicity of the parameter set.

Unary length is essential. The standard WS1S encoding of Presburger integers uses their binary representations [17, printed pp. 11–12]. Binary regularity does not imply eventual periodicity of the represented integers: the regular language 10^* represents powers of two. Our reduction uses n as unary word length, not the binary digits of t. A separate formula for each t would also be insufficient.

Boundary of the contribution. Within the literature comparison of the cited theorem statements and sections, no prior result giving the full claim from the same geometric hypotheses has been located. The priority comparison remains open. The remaining candidate contribution is the uniform, two-way translation of geometric passages, seam ownership, legal truncation and all surface-vertex collisions into one fixed MSO sentence, even with growing widths and passage counts. Fixed-width theory and the known logical conclusions do not supply that translation by themselves; we do not prove that a shorter existing reduction is impossible. The convex-face reduction is the result proved

here. The full comparison with catalogue and zigzag theory and the extension's geometric transfer obligations remain separate questions.

The two-interval arithmetic kernel is covered by Li–Chen's uniform bound for continued fractions of ratios of parameter polynomials [8, Theorem 27]. For several intervals, finite residue branches and a uniform compressed node count, the literature comparison has not located an identical general theorem; the comparison remains open.

K3's variable-modulus behaviour obstructs a single ordinary multivariable Presburger description in fixed common coordinates. Moving to canonical representatives or allowing variable divisibility does not follow from raywise periodicity. These stronger global-definability questions remain separate. General E10 $r = 1$ necessity, untranscribed families, all degenerations, human proof peer review, and a formal geometric-to-program pipeline are not completed here. A3 minimum, density and the IPR census belong to the first article. The D12 classification in Section 12 is Lean-proved.

12. A closed classification for the D12 family

The D12 member of Corollary 1b also admits a closed condition. We use the positive integer parameters of the catalogue [3]: put $L = r + 2s$, so the carbon order is $12L^2$. One quotient convention takes two chart-tagged copies of

$$H_L = \{(x, y) \in \mathbb{Z}^2 : -L \le x \le L, \quad 0 \le y \le 2L, \quad 0 \le x + y \le 2L\}.$$

In counterclockwise order its corners are $(0, 0), (L, 0), (L, L), (0, 2L), (-L, 2L), (-L, L)$. Each side is marked at distance $r+s$ from its initial corner. The alternating corners and marks have cyclic labels $(11, 0, 1, \ldots, 10)$ in the first chart and $(0, 11, 10, \ldots, 1)$ in the second. Equally labelled segments are glued in reverse order, preserving distance. Quotient vertices are chart sites modulo these identifications; adjacency is a unit lattice edge in a common chart.

Classification. For this positive-parameter quotient convention, a literal path exists if and only if

$$\boxed{r \mid (s-1) \quad \text{or} \quad r \mid (s+1) \quad \text{or} \quad |r - 3s| = 1.}$$

Equivalently, the condition is

$$r = 1 \quad \text{or} \quad (r = 2 \text{ and } s \text{ odd}) \quad \text{or} \quad \big(r \ge 3 \text{ and } (s \equiv \pm 1 \pmod r \text{ or } |r - 3s| = 1)\big).$$

Divisibility by one is automatic; modulo two the two residues both mean odd s, and $|2-3s| = 1$ adds only $s = 1$. This classification, and eventual periodicity of path existence on every positive-natural ray, are Lean-proved for the explicit quotient model above, with no prose step (Section 12.1). Its identification with the catalogue family uses the stated coordinate conventions. Two further proofs follow. They were developed separately from each other; each is computer-assisted, and its all-parameter quotient geometry, the semantics of its computations and its inductions are prose premises. Complete arguments, explicit strip formulas, certificate inputs and their checkers accompany the D12 supplement.

12.1. **A formal proof.** The Lean proof works with the quotient model directly. The two charts are glued along one boundary circle of length $6L$, and a seam vertex is recorded by its arc length. For all $r, s \ge 1$ the endpoints of an edge have at most two common neighbours and the twelve labelled points have degree five; for $r, s \ge 2$ they are the only vertices of degree five. A rotation of both charts by 60° and an exchange of the two charts are automorphisms of the quotient, so every start can be moved to the label-0 point, from which ten first triangles remain.

Necessity concerns $r, s \ge 2$, since the condition always holds when $r = 1$ or $s = 1$. For each of the ten starts the forced strip is shown to repeat a vertex before a legal endpoint whenever the condition fails. Six starts, and the other four when $s \le r + 1$, are settled by a finite cover of the negative parameter region by integer polyhedra; on each piece the strip has one fixed combinatorial form. For the other four starts with $s \ge r + 2$ the strip winds around the sphere. Writing $s = qr + c + t$ with $0 \le t < r$ and a fixed offset c, the strip consists of an initial block, a number of laps of four

blocks each, and a tail. Each lap is affine in r, s and a lap offset, so induction on the number of laps carries the strip to the tail. The repetition occurs between the last lap and the tail unless $s \equiv \pm 1 \pmod{r}$.

Sufficiency exhibits explicit literal paths: single blocks for $r = 1$ and for $s = 1$, a fixed number of blocks on $r = 3s \pm 1$ and on $s = r \pm 1$, and lap-periodic paths for $s = qr \pm 1$, $q \geq 2$, whose vertices are shown distinct for every number of laps. The ray statement follows from the classification by the arithmetic at the end of this section. The two-common-neighbour property is proved on the model, so this proof does not use cyclic edge-connectivity. It follows the strip decomposition of the two proofs below and is therefore not a third independent derivation; its correctness rests on the kernel check.

12.2. **A symbolic region cover and cylinder induction.** The first route reduces the twelve degree-five starting sites to ten representative oriented triangles by two affine lattice symmetries. The forced-prefix rule of Section 3 then follows each triangle until a legal endpoint or a repeated quotient vertex. An exact symbolic executor splits integer affine sign conditions; the earlier event is constant on each terminal region. Repetition takes precedence at a tie.

For $r \geq 3$, finitely many regions cover $r \geq s - 1$ and nine wedges cover the remaining starts through $s \leq 10r + 1$. Their positive leaves give the exceptional lines $r = 3s \pm 1$ and modular boundaries. The unbounded tail $s \geq 10r$ has an overlapping cover by cylinders. For its four residual starts, one lap increments an auxiliary state parameter by $2r$. A first exit lies in a window of width $2r$; the two positive exit planes differ by r and reduce to $s \equiv \pm 1 \pmod{r}$. This is induction on an arbitrary number of laps, rather than a bound on the number of face passages. Positive exits are compared with every earlier lap, including same-chart, seam and parallel collisions. The cases $r = 2$ have a finite complement to the tail, and $r = 1$ has explicit boundary paths.

This route requires faithful passage rows and seam maps, exhaustive integer sign splits, sound rational polyhedral tests, earliest-event and parity checks, and the full-history cylinder argument. In particular, its polyhedral emptiness shortcut uses pointedness; the supplied constraints must satisfy that condition. Finite graph audits support these semantics on their specified inputs and do not replace the all-parameter geometric and inductive arguments.

12.3. **Explicit strips and rational certificates.** The second route independently constructs the same quotient, after a translation of one chart, and derives explicit alternating strips

$$(f, A, d, e, n): \qquad A + \lfloor j/2 \rfloor d + (j \bmod 2)e, \qquad 0 \leq j < n.$$

Here f is a chart and n counts vertices. Unit differences $d, e, d - e$ produce consecutive triangular triples. The negative parameter regions are covered by short repeated prefixes or by an arbitrary number of strip pairs followed by a terminal repetition. The short part uses 29 explicit candidates and 46 region–start covers. The long part advances a strip offset by r and uses quotient automorphisms to prove the continuation for every repetition count. An exhibited repetition with no earlier legal endpoint suffices; it need not be the first repetition globally.

The infeasibility certificates use nonnegative rational weights w_i for constraints $f_i \geq 0$, with

$$\sum_i w_i f_i = -c, \qquad c > 0,$$

and all variable coefficients zero. These identities can be checked without trusting the search for the weights. Exhaustive integer coverage splits the negation of $f \geq 0$ as $-f - 1 \geq 0$. The geometric interpretation of the constraints and the arbitrary-strip induction remain part of the prose argument.

For sufficiency, explicit paths have $6s + 4$ vertices when $r = 1$, $4r + 6$ when $s = 1, r \geq 2$, and $56s + 12$ on $r = 3s + 1$, $s \geq 1$, or $56s - 10$ on $r = 3s - 1$, $s \geq 2$. For $s = qr \pm 1$, $r \geq 3$, $q \geq 1$, the alternating constructions have $2q(3r + 5s) + 2$ or $2q(3r + 5s)$ vertices; the positive-sign construction

extends to $r = 2$. Containment, endpoints and seam ownership are checked, and a global simplicity argument separates all even/odd rows, including arbitrarily separated strips. The witnesses allow internal degree-five vertices, as does the definition.

The necessity arguments use cyclic five-edge-connectivity of fullerenes [4] to exclude nonfacial dual triangles and justify unique continuation. The explicit positive unit triangles do not need that external theorem. The two routes have separately developed geometry and code; the record gives their derivation-independence and outcome-exposure qualifications. Human proof review and priority comparison remain open.

12.4. **Relation to the ray theorem.** The D12 closed condition is an additional classification, not an extra hypothesis of Theorem 1a. The convex-face decomposition already supplies D12 ray periodicity through that theorem and its prose correspondence. The classification also gives the ray result by elementary arithmetic; the Lean proof takes this route. For $r(t) = at + b$, $s(t) = ct + d$ on an eventually positive integer ray, positive a makes both divisibility decisions eventually constant: $r(t) \mid s(t) - \varepsilon$ implies $r(t) \mid a(d - \varepsilon) - bc$. If this fixed determinant vanishes, the ratio is c/a and integrality decides the condition. The exceptional equality $|r(t) - 3s(t)| = 1$ is eventually constant as well. For constant $r = m \geq 3$, a valid period divides $m/\gcd(m, c)$; $m = 1, 2$ have periods dividing one and two. The Lean ray statement starts at positive natural coordinates with nonnegative slopes; an eventually positive integer ray is covered by reindexing its positive tail.

Appendix A. Local polygons and zero boundaries

Machine-readable claim: `app:domains`.

Twice total polygon area is the carbon order only when the quotient is valid. The local coefficient audit gives

Signature	Side factors	Twice total area
E10	$p, r, r + s$	$12r^2 + 2s^2 + 12rs + 24pr + 12ps$
D12	$s, r + s$	$12r^2 + 48rs + 48s^2$
D14	r, s	$8r^2 + 24rs + 12s^2$
D1	p, r	$36p^2 + 48pr + 12r^2$
K3	r, s	$14r^2 + 12rs$
B2	p, r	$12p^2 + 24pr + 4r^2$
C1	a, b	$8a^2 + 16ab + 4b^2$
O2	t, d	$14t^2 + 16td + 2d^2$

The 71-chart/342-edge audit establishes local convexity, positive area and side lengths in its stated positive domain, not global quotient simplicity. With other parameters positive, E10 $s = 0$ and D12 $r = 0$ alone remain locally nondegenerate. E10 $p = 0$ or $r = 0$, K3 $r = 0$, B2 $p = 0$ or $r = 0$, C1 $a = 0$, and O2 $t = 0$ collapse some faces. D12 $s = 0$, either D14/D1 zero, K3 $s = 0$, C1 $b = 0$ and O2 $d = 0$ produce zero sides. Removing an IET zero-length label is not the same as redefining a graph after cone mergers. The omitted leapfrog-domain data remain in the extension's prospective records, not the main corollary. A3's $l = 0$ and $k = l$ are the separate A1 and A2 boundaries.

Appendix B. Auxiliary formal statements

The auxiliary results retain the following mathematical hypotheses.

> A finite graph with two common neighbours per edge, a cone predicate and a fixed oriented triangle has a first repeat at most the number of vertices; a literal path exists exactly when a legal endpoint occurs before that repeat.
>
> A Bézout witness for primitive u and $g \mid Q/\mathrm{cylHeight}(u, v)$ imply that the stated shifted displacement is a multiple of gu, with integer-division and divisibility hypotheses.
>
> A finite-list child relation with a strictly decreasing lexicographic natural rank has a finite uniform depth from each starting state.

The concrete equal-final-label removal component has such a depth.

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